



PHYSICS CH.10 — RADIOACTIVITY — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Radioactivity: Units & Measurement
2. Nuclear Fission vs Nuclear Fusion
3. Nuclear Reactor Fuels & Components
4. Types of Radiation: Properties & Uses
5. Sun & Stellar Energy Source
6. Discovery Facts & India's Nuclear Plants
7. ⚠ Traps / Key Differences Summary (Q1461–Q1470)

1. Radioactivity: Units & Measurement

- Radioactive material द्वारा emit hone wali radiation ki quantity measure karne ka conventional unit Curie hai; SI unit Becquerel (Bq) hai. [Q1461]
- Curie ka naam Marie Curie ke naam par rakha gaya hai. [Q1461]

2. Nuclear Fission vs Nuclear Fusion

- Nuclear Fission (विखंडन): Heavy nucleus (jaise Uranium, Plutonium) ko low-energy neutrons se bombard karke chhote nuclei mein todna; badi energy release hoti hai. [Q1464, Q1467]
- Fission ka use: Nuclear power plants (electricity) aur Atomic bomb mein hota hai. [Q1465, Q1466]
- Nuclear Fusion (संलयन): Do ya zyada halki nuclei (jaise Hydrogen isotopes) milke bhaari nucleus banate hain; energy release hoti hai. [Q1464, Q1467]
- Fusion ke liye ~100 million K temperature chahiye. [Q1464]
- Fusion ka use: Sun aur stars mein energy production; Hydrogen bomb mein. [Q1465, Q1466]
- Thermonuclear fusion ka example Hydrogen bomb hai — isme Hydrogen isotopes (Deuterium + Tritium) milke Helium banate hain. [Q1465]
- Nuclear bomb ke andar: Atomic bomb = Fission; Hydrogen bomb = Fusion. [Q1465, Q1466]

3. Nuclear Reactor Fuels & Components

- Nuclear reactor mein fissionable fuel U-235 (Uranium-235) use hota hai; isko ceramic pellets mein process karke fuel rods banate hain. [Q1463]
- Common nuclear fuels: Uranium-235 aur Plutonium-239. [Q1469]
- Heavy water (D₂O) — Deuterium oxide — nuclear power plants mein neutron moderator ka kaam karta hai; neutrons ko slow down karne ke liye. [Q1462]
- Heavy water ki khoj Harold C. Urey (1931) ne ki thi. [Q1462]
- Reactor vessel mein fuel rods paani mein doobi hoti hain jo coolant aur moderator dono ka kaam karti hai. [Q1463]
- Nuclear fusion ke main fuels: Deuterium aur Tritium (Hydrogen ke heavy isotopes). [Q1463]

4. Types of Radiation: Properties & Uses

- Alpha (α) radiation: Positively charged particles (He^{2+} nuclei); sabse kam penetrating power hai — shielding/paper se roka ja sakta hai. [Q1468]
- Alpha uses: Cancer treatment, smoke detectors, spacecraft power, oil industry. [Q1468]
- Beta (β) radiation: High-energy electrons/positrons; medium penetrating power; eye/bone cancer treatment mein use hota hai. [Q1468]
- Gamma (γ) radiation: Sabse chhota wavelength, sabse zyada energy; sabse zyada penetrating power; radiotherapy aur nuclear industry mein use hoti hai. [Q1468]
- Neutron: Subatomic particle jo atomic nuclei ka primary constituent hai. [Q1468]
- Penetrating power order: Alpha < Beta < Gamma. [Q1468]

5. Sun & Stellar Energy Source

- Sun aur anya stars ki energy ka source Nuclear Fusion hai — Hydrogen nuclei milke Helium banate hain. [Q1466]
- Ye burning of hydrogen/helium gas nahi hai; ye nuclear reaction hai. [Q1466]

6. Discovery Facts & India's Nuclear Plants

- Uranium (U) ka discovery Martin Klaproth ne 1789 mein kiya; atomic number 92; highest atomic weight among naturally occurring elements. [Q1470]
- Uranium atom mein 92 protons aur 92 electrons hote hain (6 valence electrons). [Q1470]
- India's nuclear power plants: Kakrapar (Gujarat, 1993), Tarapur (Maharashtra, 1969), Narora (UP, 1991), Kaiga (Karnataka, 2000). [Q1462]

7. ⚠ Traps / Key Differences Summary (Q1461–Q1470)

- ⚠ Trap: Curie conventional unit hai; Becquerel (Bq) SI unit hai — confuse mat hona. [Q1461]
- ⚠ Trap: Nuclear bomb = Fission; Hydrogen bomb = Fusion — donon ko alag samajhna. [Q1465]
- ⚠ Trap: Sun ki energy burning of hydrogen gas nahi, balki nuclear fusion hai. [Q1466]
- ⚠ Trap: U-235 fissionable fuel hai reactor ke liye; U-238 fertile material hai. [Q1463]
- ⚠ Trap: Alpha radiation ki sabse kam penetrating power hai — paper/shielding se roka ja sakta hai; Gamma sabse zyada. [Q1468]
- ⚠ Trap: Heavy water D_2O hai, normal water nahi — isko neutron moderator ke roop mein use karte hain. [Q1462]
- ⚠ Trap: Fusion ke liye ~ 100 million K temperature chahiye — ye fission se alag extreme condition hai. [Q1464]